

SalinO-Crop

See the salt before it kills the crop. AI-driven root-zone salinity forecasting that turns satellite data into plot-level, Bengali-language planting advice for coastal farmers.

Climate Resilience & Environmental Sustainability Satkhira · Khulna · Bagerhat · Barguna · Patuakhali **Team: [DIU Hustle Brigade]**

30–90 days

root-zone salinity forecast horizon

10–20 m

plot-level Sentinel-2 resolution

0 hardware

open data, no field sensors required

Feature phone

Bengali voice & SMS delivery

01 · THE PROBLEM

Salt is rising where farmers can't see it

Sea-level rise, tidal surges and reduced upstream freshwater flow are pushing salinity deeper into southern Bangladesh's farmland. Salt moves through soil layers and shallow aquifers, so a field that looked fine at sowing can turn toxic mid-season.

1

Invisible

Smallholders plant traditional varieties with no view of salt building up at root depth.

2

Too coarse

Current advisories are regional summaries, not plot-specific or forward-looking.

3

Too late

Losses are discovered after seed, labour and fertiliser are already spent.

The gap: farmers need a forecast, not a report. Knowing next season's root-zone salinity before planting changes which seed they choose.

02 · OUR SOLUTION

From satellite pixels to a planting decision

SalinO-Crop fuses open satellite imagery, hydrology and tide data into a plot-level salinity forecast, then matches it against crop tolerance thresholds to recommend what to grow.

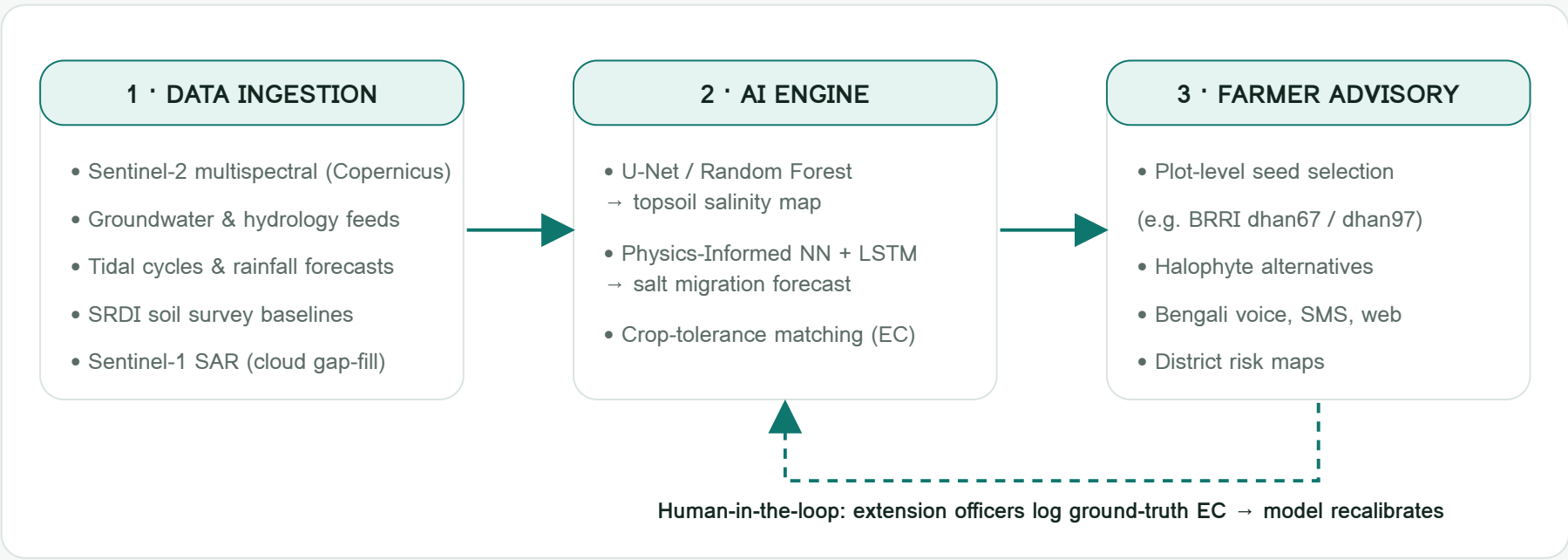


Figure 1 — End-to-end SalinO-Crop pipeline with the ground-validation feedback loop.

03 · CORE MODULES

Four capabilities, one workflow

A

Salinity surface mapping

Sentinel-2 SWIR, red and NIR bands are converted into plot-level topsoil salinity indices.

B

Salt migration forecast

A PINN + LSTM couples soil-transport physics with rainfall and tides to project root-zone salinity 30–90 days ahead.

C

Advisory engine

Predicted EC is matched against crop tolerance databases to rank suitable cultivars and halophyte cash crops.

D

Ground validation

Extension officers submit EC soil readings through a lightweight mobile web tool, continuously re-calibrating the models.

04 · TECHNICAL ARCHITECTURE

Open data, physics-aware models

Remote sensing. Sentinel-2 MSI imagery via Copernicus APIs. Salinity-sensitive SWIR bands (B11, B12) are combined with visible and NIR bands (B3, B4, B8). VNIR bands are 10 m and SWIR bands are 20 m, so products are resampled to a common grid.

Candidate index: $NDSI = (B11 - B12) / (B11 + B12)$
Index choice and thresholds are calibrated per district against SRDI and field EC data.

Machine learning. A lightweight U-Net or Random Forest separates salt crust from vegetation and surface water using multi-temporal spectral profiles. The PINN + LSTM then enforces soil-transport differential equations so forecasts stay physically plausible even where labelled data is sparse.

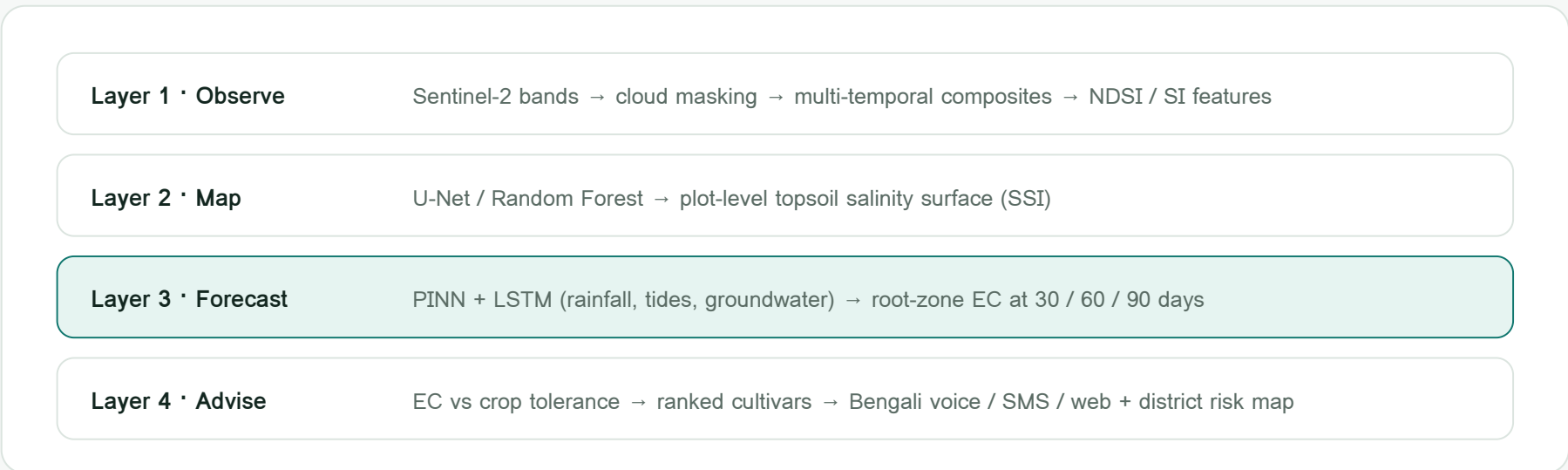


Figure 2 — Four-layer model stack. Physics-informed forecasting (Layer 3) is the core novelty.

05 · FEASIBILITY

Built to work in a village, not just a lab

Open-access foundation

Free ESA Copernicus data and public SRDI baselines keep running cost near zero.

Low-bandwidth delivery

Advisories compress into Bengali voice notes and SMS that work on feature phones.

Scalable by design

No custom hardware, so the approach can extend to all 19 coastal districts.

RISK	MITIGATION
Monsoon cloud cover blocks optical imagery	Temporal compositing plus Sentinel-1 SAR gap-filling
Satellite sees topsoil only	PINN infers root-zone salinity; validated by officer EC readings
Sparse labelled data	Physics constraints, SRDI baselines, active ground sampling
Farmer trust and literacy	Voice-first Bengali delivery through trusted extension officers

06 · IMPACT

From reacting to losses to preventing them

- **Farmers:** choose salt-tolerant varieties before planting, protecting income and inputs.
- **Communities:** secure agricultural livelihoods and reduce climate-driven distress migration.
- **Policy:** district offices get geospatial risk maps to target soil remediation and freshwater canal management.

Pilot success targets: forecast accuracy validated against field EC across pilot upazilas, advisory reach through extension officers, and a measurable share of pilot farmers changing cultivar choice based on advice. *[Insert your target numbers.]*

SalinO-Crop gives coastal farmers something they have never had: a forecast of the salt beneath their feet, and a clear answer on what to plant.